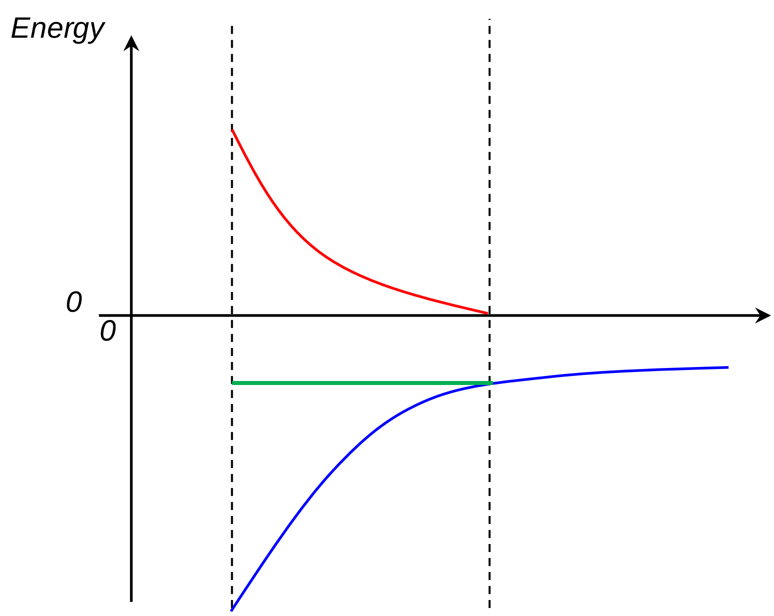
Qn 8	Answer	Marks
(a)(i)	U is inversely proportional to r.	[1]
(a)(ii)	When an external agent brings a mass from infinity to a point in a gravitational field without a change in kinetic energy, the change in gravitational potential energy is defined by the work done by this external force. Since gravitational force is <u>attractive in nature,</u> the external force acts in the opposite direction to the gravitational force to keep the kinetic energy constant. This means that the external force acts in a direction opposite to the change in displacement of mass and the work done by this force is negative. The <u>potential energy of the system when the masses are infinitely far apart is set to be zero,</u> the gravitational potential energy at that point is a negative quantity.	[1] [1] [1]
(b)(i)	 **B1** – ***E.*** ***Straight line and negative and intersect*** U ***at*** $r = r_0$ **B1 -** ***K.*** ***Curve, starts from 0.*** **B1 -** ***for relationship between the 3 graphs***	[3]

(b)(ii)	$\sum \underline{F} = m\underline{a} \Rightarrow \frac{GMm}{r^2} = ma \Rightarrow a = \frac{GM}{r^2}$ The <u>net force on the rock is the gravitational force</u> on the rock and <u>it varies inversely proportional to the square of the distance between the rock and the Earth</u>. Hence, as the rock approaches the Earth, distance between the rock and Earth decreases, hence the <u>acceleration increases</u>.	**[1]** **[1]**
(b)(iii)	$\theta = \omega t_o = \left(\frac{2\pi}{T}\right) t_o = \left(\frac{2\pi}{24 \times 3600}\right)(3600)$ $\theta = 0.2618 \approx 0.262$ rad Accept : $\pi/12$ rad.	**[1]** **[1]**
(b)(iv)	$s = R\theta = (6\,400\,000)(0.262) = 1\,680\,000$ m	**[1]**
(c)(i)	The gravitational force provides for the centripetal force for circular motion, $\frac{GMm}{(r_o)^2} = \frac{m(v_o)^2}{r_o}$ $v_o = \sqrt{\frac{GM}{r_o}}$	**[1]** **[1]**
(c)(ii)	$T = \frac{2\pi r_o}{v_o} = \frac{2\pi (r_o)^{\frac{3}{2}}}{\sqrt{GM}}$ Accept answer $T^2 = 4\pi^2 \frac{r^3}{GM}$	**[1]**

(d)(i)	$E_e < E_c$ $(U+K)_e < (U+K)_c$ $K_e < K_c$ since $U_e = U_c$ $v_e < v_c$ Since it is launched at a smaller speed and at the same distance away, the total energy of the rock undergoing the elliptical orbit will be less than the total energy of the rock undergoing circular orbit. Since at Q, (they are at the same distance away from Earth), the gravitational potential energy of the system are the same for both rocks, the kinetic energy of the rock in an elliptical orbit will be less than the kinetic energy of the rock in doing a circular orbit. The speed of rock in an elliptical orbit will therefore be less than the speed of rock when doing a circular orbit.	**[1]** **[1]** **[1]**
(d)(ii)	From Q to R, the total energy stays constant and gravitational potential energy increases. Hence, the kinetic energy decreases and hence speed decreases.	**[1]** **[1]**
	Max Marks	**20**

Qn 9	Answer	Marks